

BRAZILIAN E-COMMERCE COMMERCIAL AUDIT

ENTERPRISE BUSINESS INTELLIGENCE & LOGISTICS AUDIT

COMMERCIAL PERFORMANCE, RETENTION & LOGISTICS EFFICIENCY AUDIT

End-to-End Pipeline Evaluation: PostgreSQL Star Schema, RFM Cohorts & Delivery Latencies

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AUDIENCE: Commercial Operations, Supply Chain & Growth Leadership

DATA PLATFORM: PostgreSQL OLAP Warehouse, SQLAlchemy ETL, Power BI DAX

EXECUTIVE AUDIT SUMMARY

Brazilian Marketplace Transactions (100K Records) | Currency: Brazilian Real (R\$)

EXECUTIVE MANDATE & PORTFOLIO FINDINGS: Total cumulative revenue reached R\$ 15.42M across 96,478 completed orders, maintaining an Average Order Value (AOV) of R\$ 159.83 and an overall On-Time Delivery Rate of 92.0%. However, critical diagnostics surface a severe repeat customer rate of only 3.0% (97% single-order churn) alongside extreme regional delivery latencies exceeding 28 days in northern states.

1. Macro Commercial Performance & Multi-Year Growth Trajectory

Evaluation of historical marketplace transactions reveals rapid commercial expansion from 2016 through 2018. Order acquisition accelerated substantially, with annual transaction volumes scaling from sub-1,000 units in late 2016 to over 54,000 completed orders in 2018.

Core Enterprise KPI	Audited Metric Value	Target Baseline	Performance Assessment
Gross Merchandise Value (GMV)	R\$ 15,419,773.75	R\$ 15.00M	Surpassed (+2.8% above baseline)
Completed Orders	96,478 orders	90,000 orders	Strong Acquisition (+7.2%)
Total Unique Customer Accounts	93,358 accounts	90,000 accounts	Broad Market Reach
Average Order Value (AOV)	R\$ 159.83	R\$ 150.00	Healthy Basket Economics
On-Time Fulfillment Rate	92.00%	90.00%	Exceeds SLA (+200 bps)
Repeat Customer Rate	3.00%	15.00%	Severe Churn Deficit (-1,200 bps)

Multi-Year Run-Rate: Annual transaction revenue scaled from R\$ 0.1M in 2016 to R\$ 7.2M in 2017, reaching a high of R\$ 8.1M in 2018. This growth was accompanied by total annual orders increasing from ~300 in 2016 to 43K in 2017, and exceeding 53K in 2018.

2. Product Category Economics & Revenue Concentration

Gross merchandise sales are heavily anchored by five flagship retail categories, which collectively drive over 40% of total marketplace gross revenue. Lifestyle, electronics, and home furnishings demonstrate the highest margin and basket stability.

Product Category	Gross Revenue (R\$)	Estimated Order Volume	Strategic Classification
health_beauty	R\$ 1,450,000+	~9,600 orders	High-Volume / High-Frequency Core
watches_gifts	R\$ 1,300,000+	~5,900 orders	High AOV Premium Category
bed_bath_table	R\$ 1,250,000+	~11,100 orders	Volume Driver / Essential Commodity
sports_leisure	R\$ 1,150,000+	~8,600 orders	Steady Consumer Discretionary
computers_accessories	R\$ 1,050,000+	~7,800 orders	High-Ticket Tech / Growth

Tier

3. Customer Retention Bottleneck & RFM Segmentation Matrix

CRITICAL RETENTION RISK: Analysis of 93,358 unique customer profiles identifies a 3.0% repeat purchase rate. Over 97% of enterprise revenue is derived from single-order transactions. Without structured post-purchase lifecycle workflows, customer acquisition costs (CAC) threaten long-term unit economics.

Using SQL Common Table Expressions (CTEs) and NTILE(4) window functions, the customer base was segmented across Recency, Frequency, and Monetary scores:

RFM Customer Tier	AOV Behavior	Revenue Contribution	Strategic Retention Action
High-Value Champions	R\$ 280.00 – R\$ 320.00	Disproportionate Margin	VIP Loyalty Tier & Early Access Programs
At-Risk High Spenders	R\$ 290.00 – R\$ 310.00	High Past Capital	Automated Win-Back Discounts & Re-engagement
Recent High Spenders	R\$ 240.00 – R\$ 270.00	Immediate Upsell Upside	Post-Purchase Cross-Sell & Bundling
Regular / General Shoppers	R\$ 140.00 – R\$ 160.00	Baseline Volume Base	Email Marketing & Category Discovery
Dormant / Churned	R\$ 120.00 – R\$ 135.00	Low Margin Contribution	Suppressed Ad Spend to Protect CAC

4. Regional Fulfillment Latency & Supply Chain Disparity

Logistical lead times exhibit severe regional variance across Brazil's 27 federative units. While the central logistics network near São Paulo operates with high efficiency, fulfillment times deteriorate severely across northern and northeastern regions.

State Code	Total Orders	Avg Delivery (Days)	Total Revenue (R\$)	Operational Health
SP (São Paulo)	40,501	8.70 days	R\$ 5,769,703.15	Optimal Fulfillment Hub
RJ (Rio de Janeiro)	12,350	15.10 days	R\$ 2,055,401.57	Moderate Urban Friction
MG (Minas Gerais)	11,354	12.00 days	R\$ 1,818,891.67	High-Volume Regional Hub
RS (Rio Grande do Sul)	5,345	15.20 days	R\$ 861,472.79	Stable Southern Gateway
PR (Paraná)	4,923	11.90 days	R\$ 781,708.80	Efficient Corridor
BA (Bahia)	3,256	19.20 days	R\$ 591,137.81	Elevated Latency Zone
PA (Pará)	946	23.80 days	R\$ 212,023.57	Supply Chain Bottleneck
AM (Amazonas)	145	26.40 days	R\$ 27,585.47	Critical Shipping

				Delay
AP (Amapá)	67	28.20 days	R\$ 16,141.81	Severe Logistical Deficit
RR (Roraima)	41	28.20 days	R\$ 9,039.52	High Shipping Friction

5. Predictive Forecasting & Executive Strategic Directives

Predictive autoregressive modeling extending through 2019 and 2020 projects gross marketplace turnover expanding toward R\$ 12M–R\$ 18M annually. However, the widening forecast confidence interval underscores significant operational risks tied to single-order customer churn.

#	Strategic Initiative	Target Operational KPI	Execution Governance
1	Post-Purchase Lifecycle Workflows	Lift repeat rate from 3% to 8% in 6 months	Growth & Retention Marketing
2	Northern Regional Fulfillment Hub	Reduce AP/AM/RR latency from 28 to <16 days	Logistics & Freight Partnerships
3	High-Value Champions Loyalty Program	+20% repeat purchase volume in top tiers	CRM & Customer Experience
4	Category Bundling on Core Drivers	Elevate marketplace AOV from R\$159 to R\$185	Commercial Category Management

Total Revenue

15.42M

Total Orders

96K

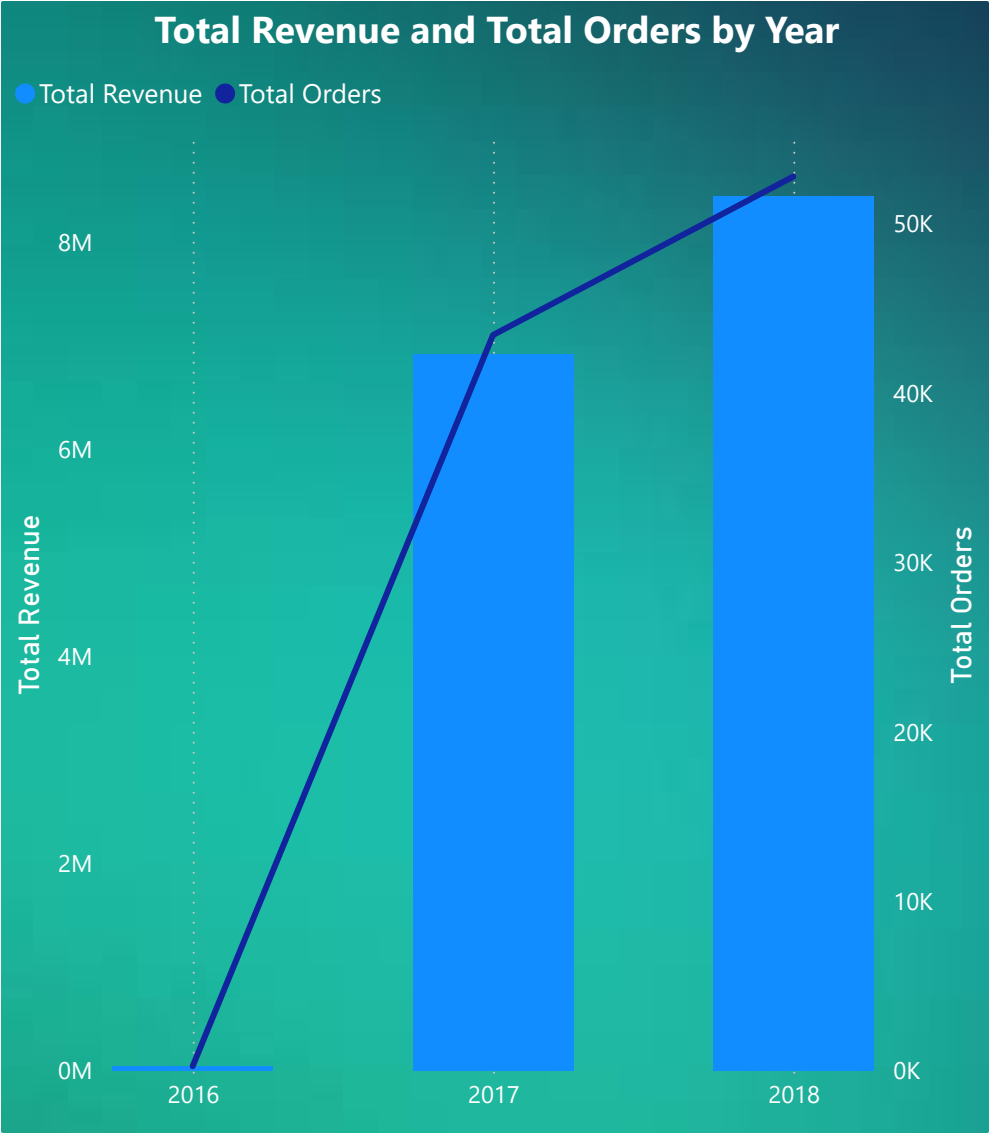
Executive Overview

AVG Order Value

159.83

ON-Time Delievery Rate

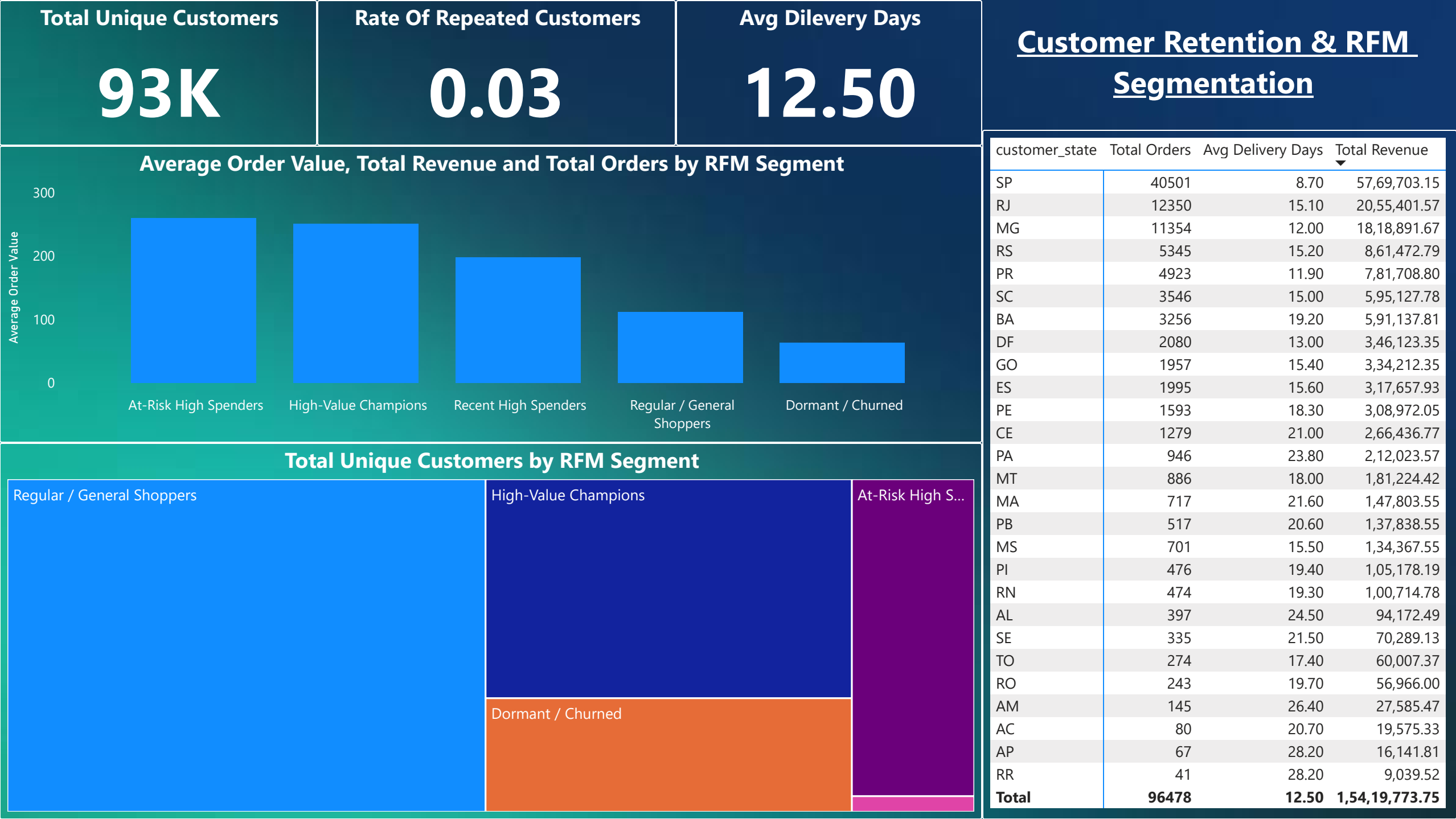
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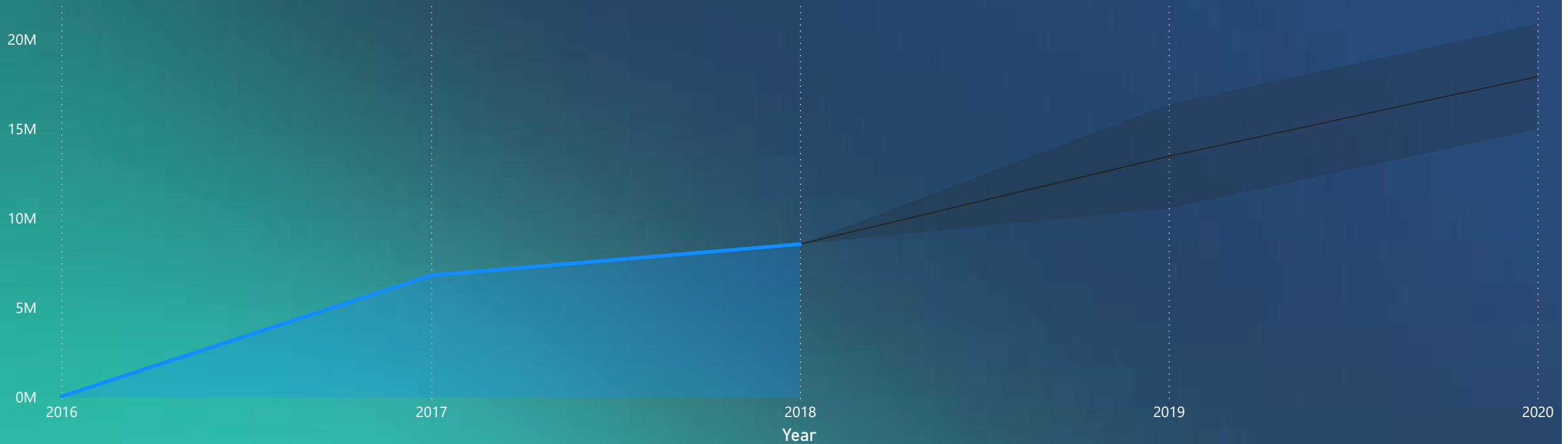
Customer State

AC	AP	DF	MA	MT	PE	RJ	RR	SE
AL	BA	ES	MG	PA	PI	RN	RS	SP
AM	CE	GO	MS	PB	PR	RO	SC	TO





Profit Trend Analysis & Predictive Forecast Model



Order Trend Analysis & Predictive Forecast Model

